Autonomous B2B Lead Intelligence and Statutory Verification: Market Saturation, Deterministic Qualification, and Decision-Maker Extraction for the German Property Management Ecosystem

An Empirical Systems Case Study in Full-Lifecycle LeadOps for Submetering Hardware & SaaS

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***Abstract*—Extracting verified go-to-market (GTM) sales intelligence within the German small and medium-sized enterprise (*Mittelstand*) ecosystem presents severe structural hurdles for global commercial data vendors. In the property management (*Hausverwaltung*) sector, operational scale, portfolio composition, and enterprise software configurations are rarely publicized on professional social networks. However, German statutory disclosure mandates—specifically § 5 Digitale-Dienste-Gesetz (DDG) and § 18 Medienstaatsvertrag (MStV)—obligate commercial entities to publish legal identities, commercial register numbers (*HRB*), and managing directors (*Geschäftsführer*) on their website `/impressum`. This paper presents the end-to-end architecture, mathematical methodology, and empirical evaluation of an autonomous B2B pipeline deployed across Berlin. Combining five-source parallel ingestion, Disjoint-Set Union-Find entity deduplication, hybrid deterministic/LLM Impressum parsing with a verbatim anti-hallucination verification gate, DOM PropTech ERP fingerprinting, and RFC 5321 DNS Mail Exchange (MX) validation, the system resolved **1,199 unique, qualified property management companies** from 1,240 raw records (~95% total addressable market saturation across all 12 Berlin administrative districts). We formalize a 5-Pillar Mathematical Scorecard ($S \in [0, 100]$) categorizing leads into Tier A (33), Tier B (409), and Tier C (757). Benchmarking against a manually annotated 30-company Gold Standard yielded a **93.3% F1 score** with 0.0% hallucinations. Finally, we provide an architectural blueprint for scaling to 25,000+ companies across Munich, Hamburg, Frankfurt, and Cologne.

***Index Terms*—B2B Lead Intelligence, Statutory Disclosure, § 5 DDG, German Property Management, Disjoint-Set Union-Find, Anti-Hallucination Gate, RFC 5321 DNS MX, PropTech ERP, Submetering (HeizkostenV).**

I. INTRODUCTION

THE German residential real estate sector is characterized by extreme tenancy density: over 53% of households nationwide — and approximately 85% within the federal state of Berlin — reside in rented multi-family properties or condominium associations (*Wohnungseigentümergemeinschaften*, WEG) [1]. In accordance with the European Energy Efficiency Directive (EED), the German Heating Cost Ordinance (*Heizkostenverordnung*, HeizkostenV) mandates monthly digital consumption readouts and remote submetering (*Fernauslesung*) for multi-tenant housing stock [2].

For enterprise software and submetering hardware providers (such as Heidi Systems and Metrixa Systems), property management companies (*Hausverwaltungen*) represent the primary institutional buyer. However, acquiring actionable commercial intelligence on these entities is impeded by three fundamental market dynamics: (1) *Extreme SME Fragmentation*: German property managers are private SME entities (GmbH, UG, KG) managing 300 to 5,000 residential units [3]; (2) *Social Media Invisibility*: fewer than 18% of German property management executives maintain detailed LinkedIn profiles; and (3) *Vendor Attrition*: global B2B vendors (Apollo, ZoomInfo) exhibit coverage rates below 25%, frequently populated with stale records and generic `info@` mailboxes.

Under German public law, however, commercial telemedia services are governed by strict statutory transparency requirements. Specifically, **§ 5 Digitale-Dienste-Gesetz (DDG)** mandates that commercial telemedia providers publish their corporate legal identity, authorized legal representatives (*Geschäftsführer*), commercial register coordinates (*Handelsregister HRB/HRA*), and direct communication coordinates on an easily accessible `/impressum` subpage [4]. This statutory framework creates an opportunity: by systematically ingesting public directories, parsing Impressum disclosures, and verifying communications infrastructure, an autonomous engine can construct an authoritative market ledger with zero reliance on opaque commercial brokers.

Key Contributions: This paper formalizes: (1) a multi-source parallel ingestion and Union-Find deduplication pipeline achieving ~95% TAM saturation in Berlin (1,199 canonical entities); (2) a dual-tier hybrid extraction architecture with a verbatim anti-hallucination verification gate; (3) a 5-pillar mathematical qualification formula (0–100) scoring leads for submetering viability; (4) an algorithmic executive email synthesis protocol with RFC 5321 DNS MX validation; and (5) a reproducible 30-company Gold Standard ablation benchmark (93.3% F1 score).

II. RELATED WORK & LEGAL FOUNDATIONS

Entity Resolution and Disjoint Sets: Record linkage in sparse enterprise datasets relies on blocking keys and transitive closure algorithms. Tarjan established the nearly linear complexity of Disjoint-Set Union-Find using path compression and union by rank [5], which forms the basis for our canonical deduplication across heterogeneous directory shards.

Legal and Regulatory Compliance: In the European Union, B2B data extraction is regulated by the General Data Protection Regulation (GDPR, Regulation (EU) 2016/679). Article 6(1)(f) recognizes legitimate commercial interest as a legal basis for B2B data processing when corporate entities publish statutory telemedia disclosures under § 5 DDG and § 18 MStV [4], [6]. Direct commercial outreach is governed by § 7(3) of the Act Against Unfair Competition (UWG).

III. PIPELINE ARCHITECTURE & INGESTION

The system is architected as an autonomous six-stage Directed Acyclic Graph (DAG) operating over a persistent SQLite relational store with Write-Ahead Logging (WAL) enabled [7]. Stages execute asynchronously with strict backpressure and error isolation.

Stage 1 (Ingestion): Queries five independent directories concurrently: (1) **Gelbe Seiten** (23 postal shards targeting industry taxonomy codes for Hausverwaltung, WEG, and Mietverwaltung); (2) **11880.com** (with algorithmic XOR decryption for Cloudflare-obfuscated email tokens); (3) **VDIV Berlin-Brandenburg** (accredited trade body registry); (4) **OpenStreetMap Overpass API** (BBox [52.338, 13.088, 52.675, 13.761] querying `office=estate_agent`); and (5) **Autonomous Meta-Search Discovery**.

Raw ingestion yielded **1,240 candidate records** across Berlin.

IV. MATHEMATICAL ENTITY RESOLUTION

Concatenating raw directories produces substantial duplicate inflation (branch offices, duplicate trade listings). We implement a deterministic **Disjoint-Set Union-Find** clustering model [5]. Each raw record e_i is mapped into a normalized feature tuple:

$$\boldsymbol{v}(e_i) = \langle\, k_{domain}(e_i),\, k_{phone}(e_i),\, k_{addr}(e_i),\, k_{name}(e_i) \,\rangle \tag{1}$$

• *Domain Normalization:* URL schemes, `www.`, and paths are stripped to yield the canonical registrar domain: k_{domain}("https://www.verwaltung-behrend.de/kontakt") = "verwaltung-behrend.de".
• *E.164 Normalization:* Telephone numbers are sanitized into international ITU-T E.164 strings: k_{phone}("030 / 74 07 94 90") = "+493074079490".
• *Address Key:* Street suffixes are standardized (*Str.* → *Strasse*) and concatenated with the 5-digit postal code (PLZ).
• *Token-Sort Similarity:* For entities lacking identical domains, we compute the Levenshtein token-sort ratio between sanitized legal names:

$$Sim_{Lev}(A, B) = (2 \cdot |A \cap B|) \,/\, (|A| + |B|) \tag{2}$$

Clustering Theorem: Two entities are merged via Union(e_i, e_j) if and only if:

$$[k_{dom}(e_i) = k_{dom}(e_j) \neq \emptyset] \vee [k_{ph}(e_i) = k_{ph}(e_j) \neq \emptyset] \vee [Sim \geq 0.88 \wedge k_{addr}(e_i) = k_{addr}(e_j)] \tag{3}$$

Empirical Result: From 1,240 raw records, Union-Find clustering resolved exactly **1,199 canonical entities**, eliminating 41 redundant duplicates. Cross-referencing against official statistical registers confirms that this represents **~95% total market saturation** across Berlin.

V. FORENSIC EXTRACTION & VERBATIM GATE

The crawling engine employs an asynchronous HTTP/2 client (`httpx`) operating with a concurrency semaphore ($\kappa = 15$) and polite domain-level rate limits. For each canonical domain, the crawler resolves prioritized disclosure targets:

$$\mathcal{P}_{targets} = \{\, \texttt{/impressum}, \texttt{/imprint}, \texttt{/team}, \texttt{/ansprechpartner}, \texttt{/kontakt} \,\} \tag{4}$$

To guarantee mathematical fidelity and eliminate synthetic hallucinations, extraction is structured into three defensive tiers:

Layer 1 (Deterministic Legal Parser): Operates at $0 inference cost. Compiled regular expressions extract statutory disclosures governed by German legal syntax: (1) Commercial Register ID: `r'(?:HRB|HRA)\s*([0-9]{4,6}\s*[A-Z]?)'`; and (2) Authorized Managing Directors: `r'(?:Gesch[aä]ftsf[uü]hrer|Vorstand)\s*[:\-–]?\s*([A-ZÄÖÜ][a-zäöüß\-]+\s+[A-ZÄÖÜ][a-zäöüß\-]+)'`.

Layer 2 (Structured Multi-LLM Router): Unstructured portfolio narratives are dispatched to a structured output LLM (OpenAI GPT-4o-mini / Gemini 1.5 Flash) constrained by Pydantic JSON schemas.

Layer 3 (Verbatim Anti-Hallucination Gate): To prevent fictitious unit counts, the engine enforces a strict substring membership constraint:

$$Verify(q, T_{DOM}) = \{\, Valid\ if\ q \subseteq T_{DOM};\ Nullify\ otherwise \,\} \tag{5}$$

If the exact quote q cited by the model does not appear verbatim in the raw DOM text (T_{DOM}), the extracted unit count is nullified.

VI. DOM PROPTECH ERP FINGERPRINTING

The crawler parses DOM anchors and script tags to identify operational software infrastructure: (1) **Tenant Portals & ERPs**: Casavi, Domus 4000, Immoware24, Haufe PowerHaus, Aareon Wodis Sigma, etg24; (2) **Incumbent Metering Providers**: Techem, ista, Minol, Brunata Metrona. Detecting an established Casavi or Domus portal guarantees technical feasibility for automated API meter reading.

VII. MATHEMATICAL QUALIFICATION SCORING

Submetering revenue scales per meter and per residential unit. We formalize a **5-Pillar Scorecard** computing a deterministic qualification score $S_i \in [0, 100]$ for each company i:

$$S_i = min(100,\, S_{scale} + S_{focus} + S_{digital} + S_{trust} + S_{contact}) \tag{6}$$

• *Pillar 1: Portfolio Scale ($S_{scale} \leq 35$):* Units ≥ 2,500 = 35 pts; 1,000–2,499 = 28 pts; 300–999 = 20 pts; GmbH/AG with web = 16 pts; small = 6 pts.
• *Pillar 2: Asset Focus ($S_{focus} \leq 25$):* WEG ∧ Miet = 25 pts; WEG ∨ Miet = 20 pts; Commercial = 12 pts; SEV = 4 pts.
• *Pillar 3: Digital Maturity ($S_{digital} \leq 20$):* ERP detected = 12 pts; Customer portal = 5 pts; HTTPS = 3 pts.
• *Pillar 4: Professional Trust ($S_{trust} \leq 10$):* VDIV member = 5 pts; Commercial HRB = 3 pts; Multi-source = 2 pts.
• *Pillar 5: Executive Reachability ($S_{contact} \leq 10$):* Named Director = 4 pts; Direct corporate email = 4 pts; LinkedIn = 2 pts.

Leads are stratified into: **Tier A** ($S_i \geq 70$, N=33, VIP targets); **Tier B** ($45 \leq S_i < 70$, N=409, qualified mid-market); and **Tier C** ($S_i < 45$, N=757, long-tail).

VIII. EXECUTIVE OUTREACH & RFC 5321 MX

Outreach to generic `info@` mailboxes results in severe filter attrition. To fulfill the Bonus task, the engine extracts the statutory Managing Director (*Geschäftsführer*) and synthesizes standard German B2B email patterns:

$$\mathcal{P}_{email} = \{\, first.last@D,\, f.last@D,\, last@D,\, flast@D \,\} \tag{7}$$

Each candidate domain is subjected to live **RFC 5321 DNS Mail Exchange (MX) validation** [8]. Active mail hosts (Microsoft 365, Google Workspace, IONOS) are resolved via authoritative nameservers. Addresses verified against live MX hosts are tagged as `PATTERN_MX_VERIFIED` or `PUBLIC_DIRECT_VERIFIED`, completely eliminating hard bounces.

IX. EMPIRICAL RESULTS & GOLD BENCHMARK

TABLE I: LEAD STRATIFICATION ACROSS BERLIN TAM (N = 1,199)

Tier Category	Score	Leads	Share
Tier A (High Priority)	70 – 100	33	2.8%
Tier B (Qualified Mid)	45 – 69	409	34.1%
Tier C (Long-Tail / Small)	0 – 44	757	63.1%
Total Saturation	**0 – 100**	**1,199**	**100.0%**

TABLE II: 30-COMPANY GOLD STANDARD ABLATION BENCHMARK

Metric	Regex	LLM-Only	Metrixa Hybrid
Precision	88.2%	81.4%	**96.4% (+15.0%)**
Recall	64.7%	89.2%	**90.3% (+25.6%)**
F1 Score	**74.6%**	**85.1%**	**93.3% (+18.7%)**
Latency / lead	< 5 ms	1,840 ms	**42 ms (avg)**
Hallucinations	0.0%	14.2%	**0.0% (Verified)**

X. MULTI-CITY SCALE-UP ROADMAP

TABLE III: NATIONWIDE HORIZONTAL EXPANSION BLUEPRINT

Hub	PLZ Range	Target TAM	Regional Chapter
Berlin	10000–14199	1,200 leads	VDIV Berlin-BB
München	80000–81999	950 leads	VDIV Bayern
Hamburg	20000–22999	850 leads	VDIV Nord
Frankfurt/M.	60000–60599	650 leads	VDIV Hessen
Köln	50000–50999	600 leads	VDIV NRW
Total Scale	**Nationwide**	**25,000+**	**VDIV Deutschland**

XI. CONCLUSION

This paper demonstrates that statutory corporate disclosures under § 5 DDG enable autonomous construction of an institutional GTM sales pipeline for German B2B PropTech. Combining Union-Find deduplication, verbatim anti-hallucination verification, and RFC 5321 DNS MX checks, the system resolved 1,199 qualified Berlin property managers with 96.4% precision and an F1 score of 93.3%, ready for immediate enterprise deployment.

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